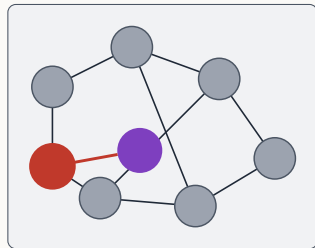


Adversarial Perturbations in Social Networks

How small graph edits can change
perceived power and influence

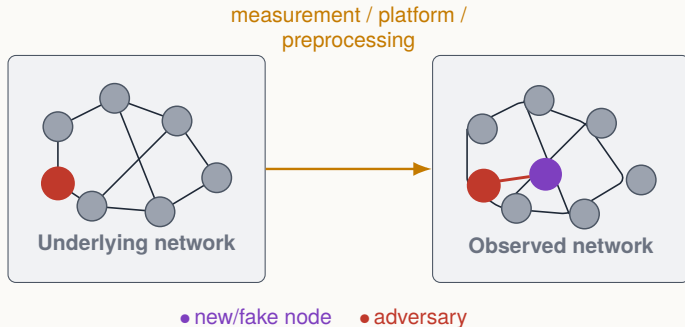
Krzysztof Czerwionka Przemysław Fuchs
TGS, MIM UW

Paper: Avram, Mishra, Parulian, Diesner (ASONAM 2019)
Class: game-theoretic approach to social network analysis



red = adversary purple = fake node
dashed red = added/removed tie

We rarely analyze the real social network



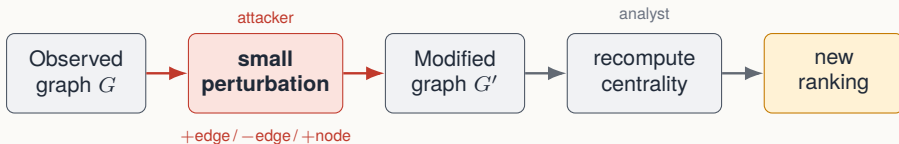
Observed $\neq$ underlying

Real ties may be hidden, missing, or distorted.

Where the attack hides

Centrality runs on the observed graph — manipulating observation manipulates the ranking.

Can a node manipulate how influential it appears?



Boost me

bottom 10% → top 10%

look more influential than I am

Hide me

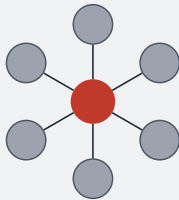
top 10% → bottom 10%

stay out of the spotlight

Goal Not to change reality — to change *perceived* power and influence.

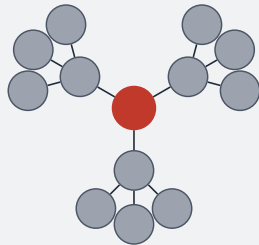
What does “influential” mean? — local view

Degree



How many direct connections?

Closeness

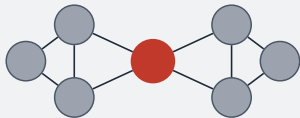


How short are paths to everyone else?

Both look at the focal node and its neighborhood.

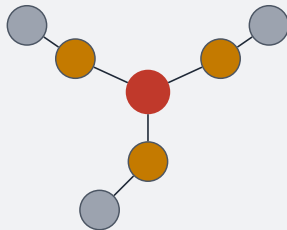
What does “influential” mean? — structural view

Betweenness



How often is it on shortest paths?

Eigenvector



Is it connected to influential nodes?

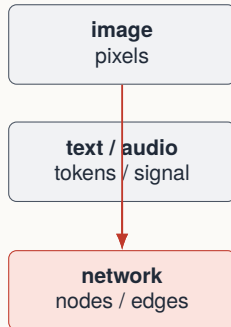
orange = already influential

Both depend on the node's position in the wider graph.

The paper borrows the logic of adversarial attacks

Domain	Small change	Wrong outcome
Image	pixel noise / sticker	classifier flips label
Text/audio	small wording or signal change	model changes prediction
Network	add/remove nodes or edges	centrality rank changes

Same idea tiny input change → large output change.
Here: graph edits → changed centrality ranking.



What exactly do Avram et al. study?

Goal

Change one node's centrality rank.

Moves

Add/remove nodes or edges — locally or globally.

Constraint

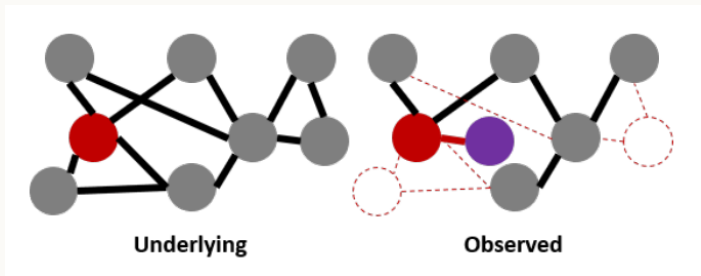
A single cheap move, cost = 1.

Evaluation

Can the rank jump between bottom and top 10%?

Setup four topologies $\times$ four centrality measures — controlled simulation of **which moves work**, not a real-platform attack.

Example: manipulating the observed network



Avram et al. (2019), Figure 1: underlying vs. observed network.

Add fake node

Create an account not present in the real network.

Hide / remove tie

Make an existing edge invisible.

Prevent observation

Keep nodes or ties out of the dataset.

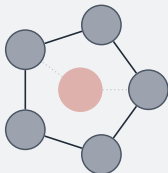
The attacker is not breaking the algorithm; they are editing its input.

Earlier work focused mostly on hiding influence

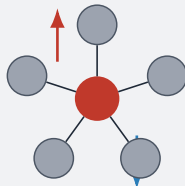
Aspect	Waniek, Michalak et al. (2018)	This paper
Goal	hide node / community	hide <i>or</i> boost a node
Changes	local edge changes	local + global; nodes + edges
Success	ranking decreases	top $\leftrightarrow$ bottom 10%
Search	multi-round heuristic	exhaustive cheap-move search

Avram et al. extend prior *node hiding* into a two-sided framework: hide *or* boost.

Two regimes side-by-side



Waniek, Michalak et al. (2018)
red node fades out



This paper (Avram et al. 2019)
red node can hide *or* become prominent

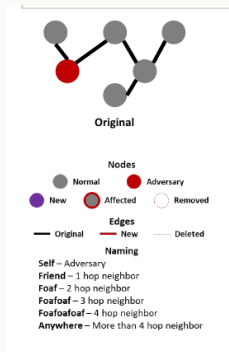
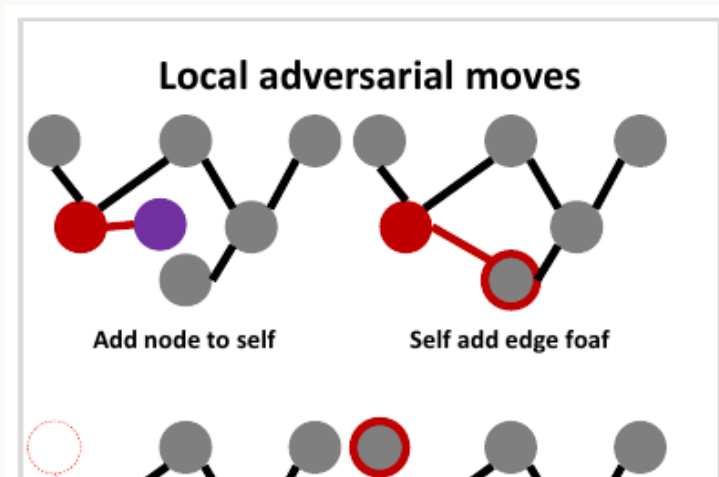
Hiding only

Adversary wants to disappear from the top of the ranking.

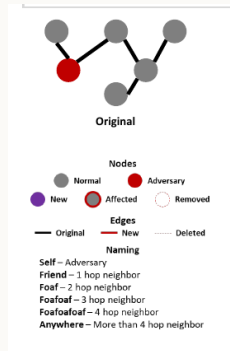
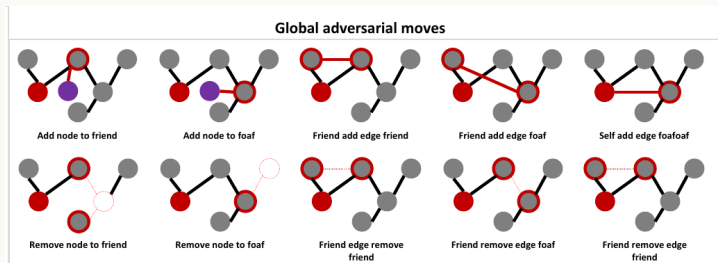
Hide or boost

Adversary can move *either* direction — down out of the top, or up from the bottom.

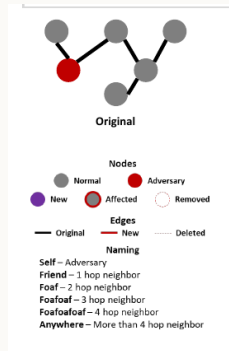
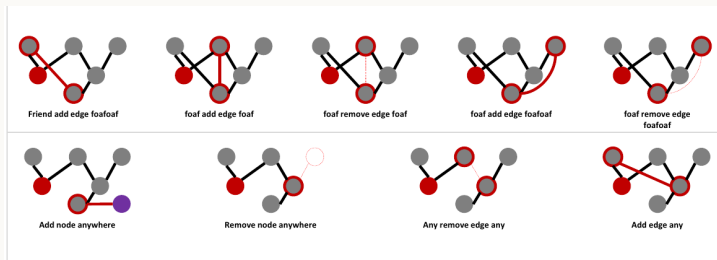
Local adversarial moves: edits in the ego network



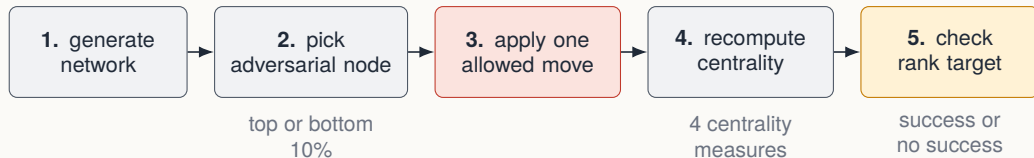
Global adversarial moves: edits anywhere in the graph (1/2)



Global adversarial moves: edits anywhere in the graph (2/2)



Experimental loop: try a move, recompute, check success



Greedy exhaustive search

Try *all* allowed single moves. Keep the move that best optimizes the target.

Handoff to Speaker B

Which networks did they test, and which moves actually worked?

The experiment grid

Dimension	Values
Networks	random, scale-free, small-world, cellular
Centralities	degree, closeness, betweenness, eigen-vector
Direction	increase or decrease
Reach	local or global

Main question Is one cheap move enough to flip the rank?

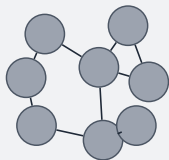
100

networks per topology

20-node graphs; cellular =
100 nodes built from five
random clusters.

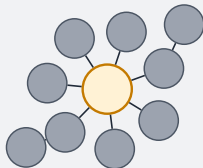
They test four common network topologies

Random



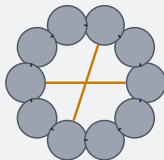
edge prob. $p = 0.2$

Scale-free



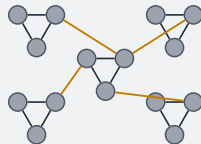
hub structure

Small-world



ring + shortcuts

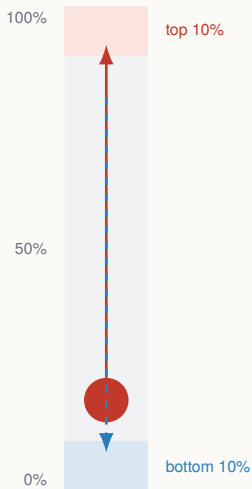
Cellular



5 clusters + bridges

Same attack, different topology — different optimal move.

Success means jumping between extremes



Increase attack

bottom 10% $\rightarrow$ top 10%

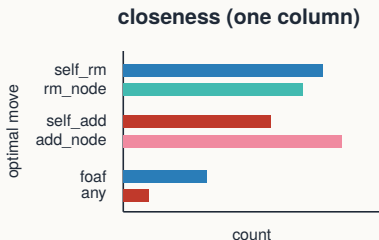
Decrease attack

top 10% $\rightarrow$ bottom 10%

Not a small score nudge

The paper measures whether a single move causes a *drastic rank movement* between extremes.

Before results: how to read Figures 3–7



Color legend

- decrease_global
- decrease_local
- increase_global
- increase_local

Reading the plot

- longer bar = move chosen more often
- columns = different centralities
- figures (3–6) = different topologies

To hide, the adversary mostly cuts itself off

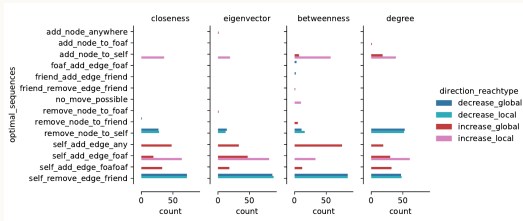
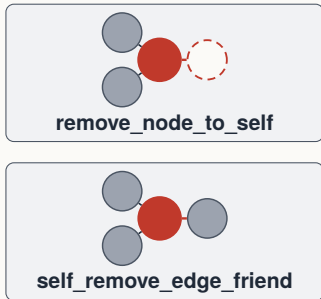


Fig. 3 — random networks. Blue / teal bars are decrease attacks; note the long bars on self_remove_edge_friend.

Hiding has a simple recipe Drop the most informative tie or neighbor; same two moves dominate across topologies and centralities. Local moves are often enough.

To look more powerful, the best move depends on context

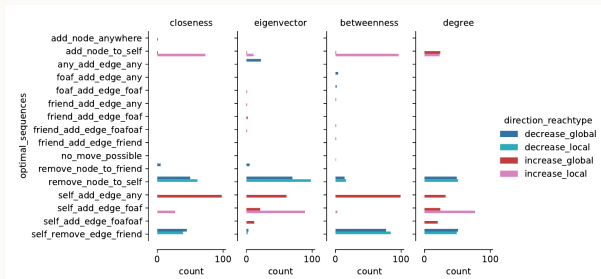


Fig. 6 — cellular networks. Red / pink bars spread across many moves; the best one shifts per centrality.

Boosting is varied; hiding is concentrated. Global reach matters more for boosting.

add_edge_anywhere

Reach beyond the ego network.

self_add_edge_foaf

Connect to a friend-of-friend.

add_node_to_self

Attach a brand-new node.

Global moves matter more for boosting than hiding

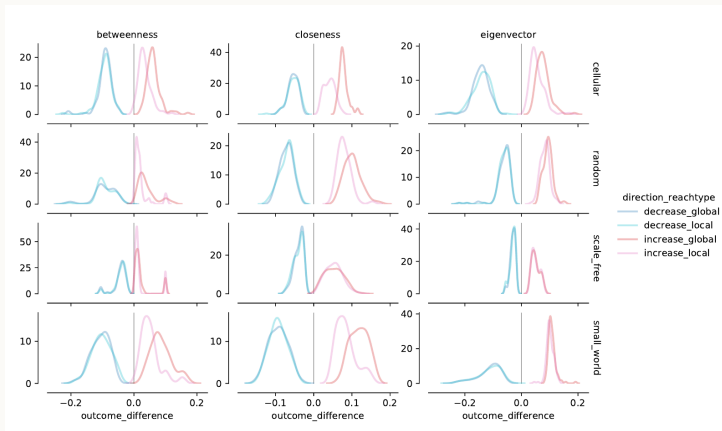


Fig. 7 — distribution of centrality change. Left of 0 = decrease, right of 0 =

Decrease

Local and global overlap — one cut is often enough.

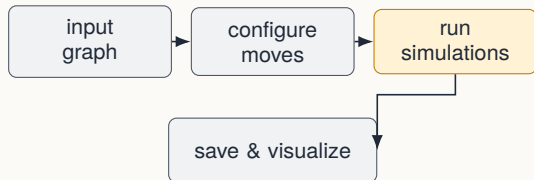
Increase

Global moves create *larger* gains for closeness.

Degree omitted

Changes are ± 1 — trivial distribution.

They also built a simulation tool



Built on Python + NetworkX.

What the tool does

- generate synthetic networks
- import real-world networks
- configure perturbation criteria
- search possible graph changes
- save and visualize results

Framed as a reusable framework rather than a one-off experiment.

What should we be careful about?

1. Scale

Mostly small synthetic networks (20–100 nodes). Exhaustive search is hard on large graphs.

2. Realism

Some global moves may be socially expensive or impossible.
Fake nodes and edge deletions have very different real-world costs.

3. Model scope

Single adversarial node.
Mostly undirected, unweighted, static graphs.
Real platforms are temporal and directed.

Open question Would the same strategies work on real social media data, or would platform constraints absorb them?

Influence metrics are attack surfaces

Small graph edits can cause large changes in perceived power.

1.

Observed networks are fragile.

2.

Centrality rankings can be gamed.

3.

Robustness should be part of SNA.

Discussion

If centrality can be flipped by a single move, should platforms or researchers report it as a *stable* measure of influence?

